



Borrowed size and agglomeration shadows in Japan: a spatial econometric analysis of regional productivity

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Abstract

This study examined the influence of spatial interdependencies on regional total factor productivity in Japan by focusing on the contrasting effects of borrowed size and agglomeration shadows. Using panel data from 47 Japanese prefectures between 2000 and 2018, we compared the manufacturing and non-manufacturing sectors by estimating the Spatial Durbin Model and the Spatial Durbin–SARAR Model. These models enable a detailed decomposition of productivity effects into direct (within region) and indirect (spillover) components across space. The results indicated that industrial agglomeration and transport capital positively influence local productivity in both sectors. However, negative spillovers, interpreted as agglomeration shadow effects, were also observed, where the concentration in neighboring regions suppressed local performance. Human capital exerted a consistently positive impact on manufacturing, with evidence of interregional spillover effects. These findings highlight the sector-specific spatial dynamics and indicate the need for differentiating regional policy responses. For the manufacturing sector, promoting networked industrial clustering and labor mobility can enhance productivity but must be balanced against the risks of overconcentration. In the non-manufacturing sector, decentralized service economies and spatially equitable infrastructure investments are critical for reducing competitive displacement from urban cores. By clarifying the conditions under which borrowed size and agglomeration shadow effects prevail, this study contributes to spatial productivity theory and offers actionable insights for designing regionally differentiated development strategies.

Keywords Total factor productivity · Agglomeration shadow · Borrowed size · Spatial econometrics · Japan

JEL Classification R11 · R12 · C21 · O47 · R15

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1 Introduction

This study investigates the influence of interregional spillover effects on regional total factor productivity (RTFP). Two contrasting spatial mechanisms mediate these effects: borrowed size and agglomeration shadows.

Borrowed size refers to the phenomenon in which a region experiences productivity gains by leveraging the external influence of economic resources available in neighboring regions, such as industrial agglomeration, human capital, and transport capital. These three elements are conceptualized as the core components of borrowed size, and their spatial spillover effects on RTFP are empirically examined using spatial econometric models.

Rooted in the theory of external economies introduced by Alonso (1973, p. 200), Burger et al. (2015) further developed the concept of borrowed size. This view is based on the premise that small cities can transcend their physical constraints and achieve growth by leveraging the economic benefits of nearby large cities, including population size, critical market mass, and functional diversity. This perspective suggests that agglomeration economies can transcend administrative boundaries into broader regional spaces (Parr 2002).

This outlook provides a theoretical foundation for urban growth mechanisms that do not rely on conventional factors such as population size or administrative status. Several studies have shown that extensive interregional networks and infrastructure can enhance functional specialization and increase productivity, overcoming the limitations of geography (Camagni 1993; Capello 2000; Hall and Pain 2006; Boix and Trullén 2007; Burger and Meijers 2016). Moreover, agglomeration economies have been shown to foster innovation and productivity growth, particularly in urban settings, where dense interactions promote knowledge spillovers (Carlino and Kerr 2015).

Advanced transportation infrastructure, such as high-speed rail networks and airports, can stimulate interregional knowledge exchange, thereby promoting knowledge diffusion and expanding the spatial scope of external economies (Ma and Liu 2022; Tian et al. 2022). These high-quality networks enhance the speed and frequency of knowledge flows, generating productivity gains for individual economic agents and through positive spillovers to surrounding regions.

However, interregional network formation does not always yield beneficial outcomes. As Hesse (2014) argues, network-based urban growth may strengthen the concentration of resources in large cities, creating a siphoning effect on the periphery. Echoing this concern, Dijkstra et al. (2013) suggested that interurban competition can widen growth disparities and undermine overall economic efficiency.

This dynamic was formalized in the concept of the agglomeration shadow introduced by Fujita et al. (1999). This describes a situation in which large, highly agglomerated urban centers absorb talent and resources from neighboring small- and medium-sized cities, thereby hindering their autonomous development (Fujita and Thisse 2002). Although borrowed size and agglomeration shadows seem to be opposing forces, they can operate either complementarily or

competitively, shaping the spatial dynamics of regional economies (Phelps and Ozawa 2003; Meijers 2005; Burger and Meijers 2016).

The simultaneous presence of benefits and constraints is a common phenomenon in real-world regional development, where proximity to large cities either promotes or inhibits local economic activity. This duality reflects the distinction between static and dynamic agglomeration economies, where immediate productivity gains may be accompanied by long-term structural disparities, depending on the spatial context (Camagni et al. 2015). When borrowed size effects are strong, they may offset the negative impacts of agglomeration shadows and support autonomous regional growth. Conversely, when agglomeration shadows dominate, they may restrain the borrowed size effects and exacerbate regional disparities.

However, this dual mechanism has gained scholarly attention, as empirical research on the simultaneous operation of borrowed size and agglomeration shadow effects remains limited. Most existing studies tend to analyze these mechanisms in isolation, often focusing on either positive externalities from urban networks or negative spillovers from concentrated agglomerations. This fragmented approach has hampered the comprehensive understanding of how these forces interact in space.

Despite the increasing interest in these concepts, the operationalization of borrowed size remains inconsistent and ambiguous. Recent empirical studies explored how spatial externalities and structural embeddedness shape regional economic outcomes in the context of urbanization and systemic shocks. Crescenzi et al. (2022) emphasized the role of territorial embeddedness in fostering local development, showing how region-specific assets can amplify or mediate external effects. Similarly, Capello and Caragliu (2023) examined the economic impact of COVID-19 in cross-border regions, revealing the uneven spatial vulnerability of regions and the importance of structural conditions in regional resilience. Guo (2025) offered a novel perspective on urban productivity by investigating the relationship between urban wage premiums and spatial equilibrium, contributing to a broader understanding of scale effects and agglomeration benefits.

These recent contributions reinforce the relevance of analyzing positive and negative spatial spillovers in productivity, particularly through frameworks that capture complex spatial interdependencies, such as those employed in this study. Some studies equate this with functional connectivity or urban performance (Meijers and Burger 2015; Meijers et al. 2015; Volkmann and Rusche 2020), whereas others emphasize specific spillover channels, such as commuting patterns or infrastructure networks (Partridge et al. 2009; Hesse 2014). These analyses frequently adopt single-dimensional frameworks and are largely centered on European urban systems. Conversely, empirical evidence from non-Western, mature regional systems, such as Japan, is limited despite rich regional heterogeneity and established regional hierarchies. As Hornbeck et al. (2024) noted, the conditions under which borrowed size dominates agglomeration shadows remain largely unresolved.

This study addresses these gaps by employing spatial econometric models to identify the pathways through which industrial specialization, human capital, and transport capital mediate spatial spillovers. By jointly estimating the borrowed size and agglomeration shadow effects, this analysis offers a multidimensional

and contextually grounded contribution to our understanding of regional productivity dynamics in Japan.

Recent studies in regional science have clarified the roles of these three pillars in shaping local productivity. Regarding industrial specialization, de Souza Silva et al. (2024) analyzed the links between occupational specialization and wage premiums in Brazil, highlighting the productivity implications of skill matching within urban labor markets. Leitão et al. (2024) revisited the structure of Marshallian districts in Portugal, offering new methodological insights into the spatial organization of industries.

Regarding human capital, a growing body of literature emphasizes cross-border spillovers and place-specific dynamics. For instance, Jardon et al. (2024) showed how human capital accumulation interacts with administrative boundaries to shape regional performance, whereas Biedka et al. (2022) provided evidence from Poland regarding the impact of EU cohesion policy investments on local human capital and development outcomes. Raya and Resosudarmo (2024) further highlighted the institutional roots of human capital disparities, illustrating how historical legacies continue to shape regional capacity building.

Finally, recent research on transport capital focused on investment volumes and spatial impact asymmetries. Arachovas et al. (2025) examined the timing and spatial distribution of public capital spikes in Greece, underscoring the uneven benefits of infrastructure across regions. Similarly, Centurião et al. (2024) evaluated road infrastructure investments along the Latin American Integration Route, highlighting differentiated regional gains depending on the network position and accessibility profiles.

These recent contributions strengthen the conceptual underpinnings of this analysis and provide a broader empirical backdrop for understanding the structural sources of RTFP variation.

This study makes a threefold contribution to the existing literature.

First, while previous studies examined borrowed size or agglomeration shadows individually, this study is among the first to jointly identify and quantify both effects within a unified spatial econometric framework. By applying the Spatial Durbin Model (SDM) and the Spatial Durbin–SARAR Model (SDM–SARAR), the analysis disentangled the positive and negative spatial spillovers associated with industrial specialization, human capital, and transport capital, enabling the structural identification of borrowed size and agglomeration shadow effects across space.

Second, unlike studies that operationalize borrowed size through regional connectivity or functional centrality (e.g., Meijers and Burger 2015; Meijers et al. 2015; Volkmann and Rusche 2020), this study decomposes spatial externalities into three distinct channels—industrial specialization, human capital, and transport capital—to offer a granular understanding of spillover pathways.

Third, existing empirical literature focuses primarily on European or North American regional systems, whereas this study provides novel evidence from Japan, a non-Western context characterized by a high urbanization rate and polycentric regional structure. The applicability and limitations of established spatial economic theories in different institutional and geographical settings were critically assessed.

The remainder of this paper is organized as follows: Sect. 2 outlines the empirical framework, including definitions of key variables, model specifications, estimation strategies, and data. Section 3 provides the preliminary analysis, including tests for spatial autocorrelation and regional fixed effects. Section 4 presents the estimation results and discusses their interpretation and policy implications. Finally, Sect. 5 concludes the paper and suggests directions for future research.

2 Methodology

2.1 Definition of key variables

This study elucidates the influence of interregional spillover effects on RTFP, focusing on the impact of three key regional economic resources: industrial specialization (LQ), human capital (HC), and transport capital (TC). These variables were statistically examined to assess their impact on productivity.

RTFP was calculated following the methodology proposed by Good et al. (1997), as applied to regional data by Otsuka (2024a, b). The relative RTFP level of a given industry in region j at time t , denoted as $\ln\text{RTFP}_{jt}$, is defined as follows:

$$\ln\text{RTFP}_{jt} \equiv \left(\ln Y_{jt} - \overline{\ln Y_t} \right) + \sum_{s=1}^t \left(\overline{\ln Y_s} - \overline{\ln Y_{s-1}} \right) - \left[\sum_{i=1}^n \frac{1}{2} \left(S_{ijt} + \overline{S_{it}} \right) \left(\ln X_{ijt} - \overline{\ln X_{it}} \right) + \sum_{s=1}^t \sum_{i=1}^n \frac{1}{2} \left(\overline{S_{is}} + \overline{S_{is-1}} \right) \left(\overline{\ln X_{is}} - \overline{\ln X_{is-1}} \right) \right], \quad (1)$$

where Y_{jt} denotes the output of region j at time t , X_{ijt} is the input of factor i in region j at time t , and S_{ijt} is the cost share of input i . The overbars represent the interregional averages of the respective variables. This formulation enables consistent comparisons of productivity levels across regions and over time. It measures not only the relative deviations from the national average but also the dynamic evolution of regional productivity trajectories in response to national trends.

The location quotient (LQ), which measures the degree of industrial specialization in a region, was defined according to Glaeser et al. (1992)

$$\text{LQ}_{kr} \equiv \left(\frac{Y_{kr}}{Y_r} \right) / \left(\frac{Y_k}{Y} \right), \quad (2)$$

where Y_{kr} is the output of industry k in region r ; Y_r is the total output of region r ; Y_k is the national output of industry k ; and Y is the total national output. A value greater than 1 implies that the share of that industry in the regional economy surpasses the national average, indicating regional specialization. Firms in such regions benefit from localized external economies (Rosenthal and Strange 2004; Combes and Gobillon 2015). These effects can include knowledge spillovers through direct interaction and labor mobility, contributing to higher RTFP. However, high specialization

in neighboring regions may siphon away resources, creating an agglomeration shadow.

Human capital (HC) serves as a proxy for workforce education and skill levels. As no official composite index was available, the following wage-weighted measure was constructed:

$$HC_{jt} \equiv \frac{\sum_n \sum_s w_{snt} * L_{snt}}{L_{jt}}, \quad (3)$$

where w_{snt} is the relative wage for workers of gender s and education level n , L_{snt} denotes the number of such workers, and L_{jt} represents the total workforce in region j at time t . Regions with a greater proportion of highly educated workers (e.g., university graduates with above-average wages) will exhibit higher HC values. Conversely, low HC values indicate educational disadvantages. Human capital enhances productivity directly and through spillover effects (Habib et al. 2019; Ghosh and Parab 2021).

Transport capital (TC) indicates the infrastructure stock that supports economic activities, particularly transportation. In this study, it is defined as the aggregate social capital stock for roads, ports, and airports. The study data were sourced from official Japanese statistics on public capital formation. A well-developed transportation infrastructure facilitates the movement of goods, reduces costs, and improves productivity (Eberts and McMillen 1999; Bröcker et al. 2019). Such infrastructure enhances the productivity of local firms and strengthens their capacity to participate in interregional economic networks.

2.2 Strategy for model selection

To account for spatial economic interdependence, this study adopted the SDM and SDM-SARAR models. These models are suitable for analyzing regional productivity patterns where observable characteristics and unobserved spatial factors are likely to generate interregional spillovers.

Traditional spatial econometric approaches, such as the Spatial Autoregressive Model (SAR), which introduces a spatial lag of the dependent variable, and the Spatial Error Model (SEM), which incorporates spatial autocorrelation in the error term, have been widely used. However, each model captures only one aspect of spatial dependence and thus has limitations.

The SAR model is useful for modeling situations in which the RTFP in neighboring regions directly affects the local RTFP. However, it does not account for spatial spillovers that operate through explanatory variables, such as the influence of neighboring regions' human capital or industrial structures. Conversely, the SEM model addresses spatially correlated unobserved factors but does not capture the impact of the observed variables in neighboring regions.

The SDM overcomes these limitations by including the spatial lags of both dependent and explanatory variables. This structure enables simultaneous estimation of the direct (within-region) and indirect (between-region) effects of industrial specialization (LQ), human capital (HC), and transport capital (TC) on regional

productivity. Moreover, by encompassing spatially lagged regressors, the SDM nests both SAR and SEM models as special cases under specific parameter restrictions, positioning it as a generalized and robust spatial model (Anselin 1988; Elhorst 2014).

While the SDM can address the observed spatial heterogeneity, it does not fully capture the latent sources of spatial dependence. The SDM-SARAR model extends the SDM by introducing spatial autocorrelation in the error term via a spatial autoregressive parameter (λ). This addition enables the model to account for unobserved spatial dependence arising from the institutional context, cultural characteristics, or omitted geographic factors. By modeling observable and unobservable sources of spatial dependence, SDM-SARAR enhances identification and improves the consistency of parameter estimation.

Overall, the SDM and SDM-SARAR models offer the following methodological advantages over conventional spatial econometric approaches: (1) they support the explicit modeling of spatial spillover effects; (2) they jointly consider both observable and latent spatial dependence; and (3) they are general and inclusive frameworks that nest SAR and SEM models. Given these strengths, they are well suited for capturing the nuanced spatial interdependencies that shape regional productivity.

2.3 Estimation models

1. Spatial Durbin Model (SDM)

The SDM enables the explicit estimation of spatial spillover effects by incorporating the spatial lags of dependent and independent variables. This feature is useful for analyzing regional productivity, which may be influenced by local conditions and neighboring characteristics. The model is expressed as follows:

$$Y^k = \rho WY^k + X^k\beta + WX^k\theta + \mu + \varepsilon, \quad (4)$$

where Y^k is a vector of the RTFP for industry k (manufacturing or non-manufacturing), and $X^k = [LQ^k, HC, TC]$ is the matrix of explanatory variables, including location quotient (LQ), human capital (HC), and transport capital (TC). W denotes the spatial weight matrix; ρ is the spatial autoregressive coefficient; β captures the impact of local explanatory variables; θ represents the effects of neighboring regions' explanatory variables (that is, spillover effects); μ is a vector of region-specific fixed effects; and ε is the error term.

The coefficients in θ reflect the impact of “borrowed size” from the neighboring regions. A positive value ($\theta > 0$) indicates positive spillovers, namely, that external resources concentrated in nearby areas enhance local productivity. Conversely, a negative value ($\theta < 0$) implies that economic resources are drawn away toward neighboring regions, resulting in a reduction in local productivity due to an “agglomeration shadow.”

The SDM nests the SAR and SEM as special cases under specific parameter constraints (Anselin 1988), making it a general and flexible framework for capturing different types of spatial dependence (Elhorst 2014).

2. Spatial Durbin–SARAR model (SDM-SARAR)

The SDM-SARAR Model extends the SDM by incorporating spatial autocorrelation into the error term. Assuming that the error follows a spatial autoregressive process, the model is specified as follows:

$$Y^k = \rho WY^k + X^k\beta + WX^k\theta + \mu + u, u = \lambda Wu + \varepsilon, \quad (5)$$

where λ is the spatial autoregressive coefficient in the error term, capturing unobserved spatial dependence arising from institutional settings, cultural background, or unmeasured geographic features. Thus, the SDM-SARAR allows for a richer representation of spatial dependence by modeling observable spillovers (via WX) and latent dependence (via λWu).

As noted by Elhorst (2010), the SDM-SARAR model may be subject to identification issues under specific conditions. However, this study addresses these concerns by employing the full Durbin specification (that is, applying spatial lags to all explanatory variables) and using a panel dataset with sufficient spatiotemporal variation to ensure a stable estimation. The empirical performance of the model is further examined in the results section.

The SDM and SDM-SARAR models were estimated by applying maximum-likelihood (ML) methods based on prefecture-level panel data with regional fixed effects. ML has been widely adopted in spatial econometrics (Ord 1975; Haining 1990, 2003), and the balanced panel used in this study provides adequate observations for consistent estimation.

These models incorporate three common sources of spatial correlation: spatial lag of the dependent variable (WY), spatial lag of explanatory variables (WX), and spatial autocorrelation in the error term (λWu). Each of these sources can introduce endogeneity. Specifically, a spatial lag of the dependent variable may result in simultaneity bias. However, the ML estimator was designed to account for this endogeneity, and yielded consistent estimates.

Moreover, the Durbin specification facilitates the identification of spatial spillover effects through WX , whereas the SARAR structure helps control for spatially correlated omitted variables. This combination enhanced the credibility of the estimated relationships between regional productivity and spatial drivers.

In summary, estimating the SDM and SDM-SARAR using ML provides a theoretically grounded and empirically robust approach for evaluating spatial externalities. These models enable a statistically reliable analysis of how borrowed size mechanisms, industrial agglomeration, human capital, and transport infrastructure affect regional productivity across Japanese prefectures.

3. Impact evaluation

In spatial econometric models with lag structures, changes in an explanatory variable in one region can affect that region as well as other regions through spatial linkages. This feature introduces complexity in interpreting model coefficients, as the estimated parameters (such as β and θ) do not directly represent the full magnitude of influence across space.

To address this problem, LeSage and Pace (2009) proposed decomposing the estimated effects into three components: direct, indirect, and total. Currently, this approach is the standard for analyzing spatial models, such as SDM and SDM-SARAR.

The direct effect refers to the average impact of a change in a region's explanatory variable on its dependent variable (RTFP). This reflects localized mechanisms, such as agglomeration economies and internal policy interventions.

The indirect effect (that is, the spatial spillover effect) measures the average influence of changes in explanatory variables in neighboring regions on the RTFP of a given region. This component captures the interregional externalities that are central to evaluating the balance between positive and negative agglomeration shadows.

The total effect is the sum of the direct and indirect effects and represents the overall spatial impact of a given variable.

In the general SDM-SARAR framework, these effects are computed numerically using the following impact matrix:

$$\frac{\partial Y^k}{\partial X^k} = (I - \rho W)^{-1} (\beta_X I + \theta_X W), \quad (6)$$

where β_X and θ_X are the coefficients associated with the local and spatially lagged explanatory variables, respectively. The average of the diagonal elements represents the direct effect, the average of the off-diagonal elements represents the indirect effect, and the overall mean represents the total effect.

For instance, if all three effects are positive, this suggests that the variable in question, such as human capital or transport capital, not only improves local productivity but also generates positive spillovers to other regions. These findings support the presence of borrowed size effects through interregional connectivity.

Conversely, if a variable exhibits a significantly negative indirect effect, it indicates the presence of an agglomerated shadow. In this case, strong concentrations of industrial specialization (LQ), human capital (HC), or transport capital (TC) in neighboring areas drain productive resources, thereby reducing RTFP in adjacent regions despite local efforts to improve productivity.

This impact decomposition provides a rigorous analytical basis for understanding the role of uneven spatial distribution of economic resources in shaping regional development outcomes. By distinguishing between local and spillover effects, this analysis highlights the dual role of agglomeration: it can either facilitate or constrain regional growth, depending on how benefits and burdens are spatially distributed.

Thus, the SDM-SARAR model is a powerful tool for quantifying these dynamics and enables a comprehensive empirical investigation of the spatial propagation mechanisms that drive regional productivity in Japan.

2.4 Spatial weight matrix

This study constructed a binary queen-type contiguity matrix to represent the geographical proximity between Japan's prefectures (Anselin 1988). In this matrix, two prefectures are considered neighbors if they share either a common border or a single point of contact. The matrix is square and symmetric, with a value of 1 assigned to adjacent pairs and 0 otherwise. The diagonal elements are set to zero, indicating that a prefecture is not its own neighbor.

Compared to the rook-type matrix, which only includes edge-sharing units, the queen-type matrix incorporates corner contiguity, providing an inclusive representation of spatial connectivity. This is particularly relevant for Japan, where prefectural boundaries are often irregular and fragmented. The queen approach better reflects real-world spatial interactions, including broad-area effects such as resource allocation pressures and spillover mechanisms, which are the central themes of this study.

A specific adjustment was applied to Okinawa Prefecture, which is geographically isolated and does not share direct borders with the other prefectures. To avoid numerical issues caused by zero-row isolation, Okinawa is considered adjacent to Kagoshima Prefecture, which is its geographically nearest mainland counterpart. This procedure follows established practices in applied spatial econometrics (LeSage and Pace 2009; Elhorst 2014), in which spatial islands are connected to the closest unit to ensure full participation in spatial structures.

The adjacency matrix was subsequently row-standardized to form the spatial weight matrix used in the estimation models (LeSage and Pace 2009). In this process, each non-zero element is divided by the number of neighbors for the corresponding row unit, such that all row sums are equal to one. This normalization allows for comparability across regions and prevents regions with more neighbors from exerting a disproportionately large influence.

2.5 Data

This study analyzes the determinants of regional productivity in the manufacturing and non-manufacturing sectors using panel data for all 47 prefectures in Japan between 2000 and 2018. All data sources are publicly available and based on official national statistics.

The dependent variable, RTFP, was calculated using the Regional-Level Japan Industrial Productivity Database (R-JIP) published by the Research Institute of Economy, Trade, and Industry (RIETI). This database was developed to overcome the limitations of regional and industry-consistent data availability in Japan. In this study, prefecture- and industry-level RTFP were estimated under consistent definitions for the full sample period.

The analysis period ended in 2018, which is the latest year available in the R-JIP database. Other key explanatory variables, including public capital stock, were obtained from official Cabinet Office data, with the most recent available year being 2020. These limitations stem from the deliberate emphasis on research transparency

and reproducibility using reliable open datasets. Therefore, the absence of post-COVID observations reflects current institutional constraints rather than a lack of theoretical or policy relevance.

In the RTFP calculation, output (Y) was measured as the real value added. Input (X) comprises labor and capital, with labor measured in hours worked and capital as the real capital stock. Cost shares (S) were derived from labor and capital cost data included in the R-JIP.

The LQ was constructed using real value added by industry and prefecture and was sourced from the R-JIP. HC was calculated using employment and wage data disaggregated by educational attainment from the Basic Survey on Employment Structure (Statistics Bureau of Japan) and the Basic Survey on Wage Structure (Ministry of Health, Labour and Welfare). TC, which serves as a proxy for infrastructure investment, is based on the Cabinet Office's Statistics on Social Capital Stock. Table 1 summarizes the descriptive statistics for all the variables.

A comparison of the mean RTFP values indicates that manufacturing sectors exhibit higher productivity than non-manufacturing sectors, indicating pronounced technological advancement and efficiency gains. The standard deviation is also greater for manufacturing, reflecting stronger regional disparities.

The average LQ in manufacturing slightly exceeded 1.0, indicating regional concentration, whereas the non-manufacturing LQ was closer to 1.0, suggesting a uniform distribution across regions. The standard deviation of the LQ was notably higher in the manufacturing sector, reinforcing the presence of spatially clustered industrial activities.

The HC index showed relatively stable averages but substantial variation across regions. The gap between the maximum and minimum values reflects a spatial imbalance in educational attainment, with potential implications for local productivity.

TC, which represents infrastructure investment, exhibited a moderate upward trend in its mean value. However, a high standard deviation suggests uneven infrastructure development, with some regions benefiting from substantial transport investments.

Overall, the descriptive statistics reveal sectoral contrasts in productivity and significant regional imbalances in economic structure and resource allocation. The combination of high variance in RTFP and LQ in manufacturing underscores the importance of agglomeration mechanisms. These findings provide empirical motivation for examining spatial externalities, both positive (borrowed size) and negative (agglomeration shadow), in the subsequent econometric analysis.

Figures 1 and 2 present the geographic distribution of the average RTFP during the study period (2000–2018) for manufacturing and non-manufacturing sectors, respectively. While the manufacturing sector exhibits a dispersed pattern, with multiple high-RTFP regions in western and central Japan, the non-manufacturing sector shows strong centralization, particularly around Tokyo and Osaka. Peripheral areas in both maps tend to have lower RTFP levels, suggesting potential agglomeration shadow effects. These spatial differences underscore the need for sector-specific analysis in subsequent empirical modeling.

Table 1 Descriptive statistics

	Regional total factor productivity				Location quotient		Human capital	Traffic capital
	Manufacturing sector		Non-manufacturing sector		Manufacturing sector	Non-manufacturing sector		
	RTFP (index)		RTFP (index)		LQ (index)	LQ (index)	HC (index)	TC (million yen)
2000–05	Mean	0.140	−0.002	1.022	0.993	0.765	4,750,858	
	Standard deviation	0.220	0.112	0.384	0.114	0.051	3,467,150	
	Max	0.866	0.568	1.960	1.251	0.895	18,752,129	
	Min	−0.436	−0.188	0.183	0.715	0.639	1,612,596	
2006–11	Mean	0.263	−0.018	1.066	0.981	0.732	5,093,742	
	Standard deviation	0.195	0.127	0.385	0.112	0.054	3,675,046	
	Max	0.836	0.579	2.107	1.257	0.879	19,530,940	
	Min	−0.253	−0.212	0.172	0.695	0.603	1,873,695	
2012–18	Mean	0.381	−0.005	1.094	0.973	0.825	5,244,527	
	Standard deviation	0.178	0.135	0.408	0.116	0.035	3,713,090	
	Max	0.902	0.619	2.056	1.233	0.934	20,199,760	
	Min	−0.246	−0.211	0.183	0.693	0.732	1,944,639	
2000–18	Mean	0.267	−0.008	1.062	0.982	0.776	5,041,015	
	Standard deviation	0.221	0.126	0.395	0.114	0.061	3,630,888	
	Max	0.902	0.619	2.107	1.257	0.934	20,199,760	
	Min	−0.436	−0.212	0.172	0.693	0.603	1,612,596	

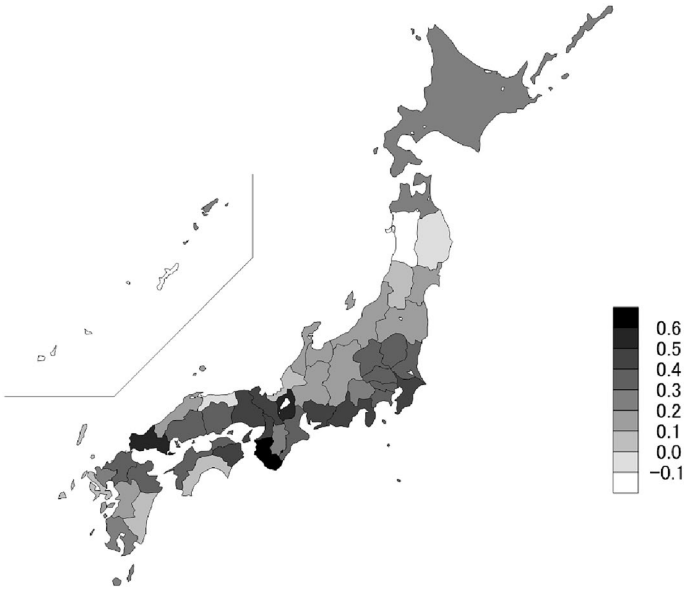


Fig. 1 Geographic distribution of average RTFP (manufacturing sector, 2000–2018)

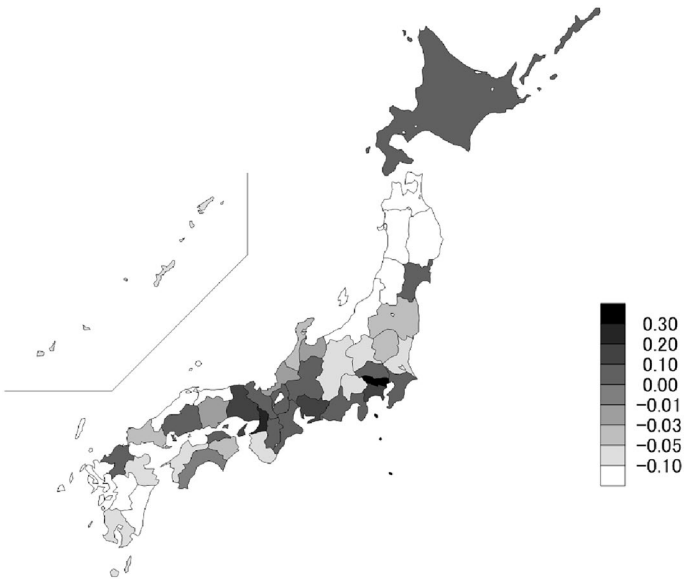


Fig. 2 Geographic distribution of average RTFP (non-manufacturing sector, 2000–2018)

3 Preliminary analysis

3.1 Spatial autocorrelation

A preliminary step in spatial econometric analysis is to assess the presence of spatial autocorrelation in the dependent variable, RTFP. This assessment clarifies whether productivity levels exhibit geographic clustering, which suggests spatial spillovers, concentration effects, or random distributions.

This study employed Moran's I (Moran 1950) as a measure of spatial autocorrelation. Moran's I is a spatial analog of Pearson's correlation coefficient and is used to measure the similarity of values among neighboring regions. This measure is particularly useful, because it is sensitive to spatial lags in the dependent variable, spatial autocorrelation in the error term, and heteroscedasticity (Anselin and Griffith 1988; Anselin and Rey 1991; Florax and de Graaff 2004).

A positive Moran's I value indicates that neighboring regions tend to exhibit similar levels of industrial activity, implying spatial clustering. Conversely, a negative

Table 2 Moran's I test results

Year	Manufacturing sector				Non-Manufacturing sector			
	Moran's I	Z score	P value		Moran's I	Z score	P value	
2000	0.45	4.41	0.00	***	0.18	2.11	0.02	**
2001	0.41	4.08	0.00	***	0.19	2.17	0.01	**
2002	0.42	4.15	0.00	***	0.22	2.47	0.01	**
2003	0.38	3.72	0.00	***	0.25	2.87	0.00	***
2004	0.36	3.51	0.00	***	0.26	2.89	0.00	***
2005	0.33	3.34	0.00	***	0.24	2.69	0.00	***
2006	0.21	2.15	0.02	**	0.26	2.93	0.00	***
2007	0.29	2.88	0.00	***	0.27	3.01	0.00	***
2008	0.27	2.71	0.00	***	0.28	3.09	0.00	***
2009	0.19	1.99	0.02	**	0.30	3.32	0.00	***
2010	0.13	1.43	0.08	*	0.34	3.66	0.00	***
2011	0.21	2.13	0.02	**	0.33	3.61	0.00	***
2012	0.18	1.89	0.03	**	0.25	2.82	0.00	***
2013	0.16	1.71	0.04	**	0.23	2.59	0.00	***
2014	0.06	0.80	0.21		0.22	2.47	0.01	**
2015	−0.02	0.00	0.50		0.20	2.23	0.01	**
2016	0.02	0.38	0.35		0.19	2.19	0.01	**
2017	0.01	0.28	0.39		0.20	2.22	0.01	**
2018	0.13	1.45	0.07	*	0.22	2.44	0.01	**
Time average	0.30	3.00	0.00	***	0.26	2.89	0.00	***

Note: statistical significance levels are indicated as follows: *** $p < 0.01$ (highly significant), ** $p < 0.05$ (strongly significant), * $p < 0.10$ (statistically significant), and no symbol ($p \geq 0.10$) indicates not significant

value suggests spatial dispersion in which adjacent areas display divergent industrial patterns. Table 2 presents the calculated Moran's I values for RTFP by sector.

In the manufacturing sector, Moran's I has remained consistently around 0.4 since 2000 and is statistically significant. This implies strong and persistent clustering of regional productivity. Although a slight decline was observed in 2003 (0.38), spatial autocorrelation overall remained stable.

In contrast, the values of Moran's I for the non-manufacturing sector range from approximately 0.18 to 0.25, indicating weaker spatial clustering. The non-manufacturing sector displays spatial dependence, and its intensity is clearly lower than that of the manufacturing sector. However, since 2003, Moran's I increased, suggesting a gradual intensification of spatial concentration.

These results indicate that the manufacturing sector consistently exhibited strong clustering, whereas the non-manufacturing sector exhibited relatively dispersed clustering. This distinction provides an important context for interpreting the borrowed size effects. In areas with strong agglomeration shadows, peripheral regions adjacent to industrial hubs may experience development constraints and negative spatial externalities in terms of RTFP. Given the statistical significance of Moran's I in both sectors, the use of spatial econometric models rather than conventional OLS is warranted.

Figure 3 illustrates the trends reported in Table 2 to explore the temporal dynamics of the spatial autocorrelation.

The economic interpretation of these patterns and the decline in Moran's I in the manufacturing sector during the early 2000s reflects a temporary weakening of interregional industrial links, potentially caused by offshoring. As capital, technology, and labor resources were relocated abroad, the spatial sharing of production inputs, which is central to the borrowed size effect, diminished. Consequently, spillover effects that enhance local productivity may have weakened,

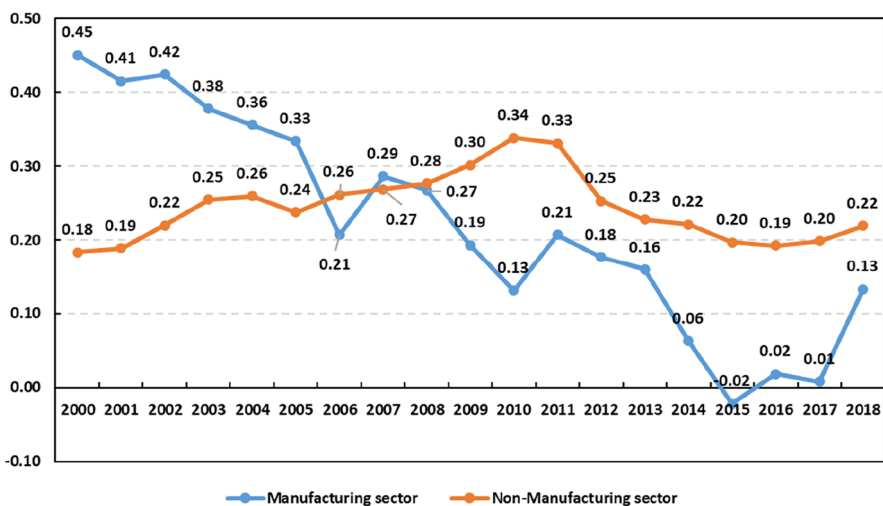


Fig. 3 Moran's I over time for manufacturing and non-manufacturing sectors

reducing spatial autocorrelation. The recovery of Moran's I between 2017 and 2018 indicates that high-value-added manufacturing activities returned to Japan, thereby revitalizing interregional linkages in capital and labor and strengthening regional industrial clusters.

In the non-manufacturing sector, the upward trend in Moran's I reflects spillover benefits from the agglomeration of advanced service industries in the Tokyo metropolitan area. Contiguous regions, such as neighboring parts of the capital region and major provincial cities, may benefit from their proximity to service hubs. Knowledge-intensive sectors, including finance, IT, and professional services, are likely to facilitate the diffusion of industrial advantages, thereby expanding service clusters to nearby areas.

These borrowed size effects encompass the benefits of neighboring agglomerations. In manufacturing, the formation of industrial clusters fosters competition, deepens labor markets, and promotes technological spillovers. The recent recovery in Moran's I suggests the reemergence of cluster-driven spillovers with the strengthening of industrial linkages. For instance, in automotive hubs such as Aichi and Hiroshima, supply chain integration and technology transfer may enhance productivity in the surrounding areas.

In non-manufacturing areas, agglomeration economies in metropolitan areas are likely to facilitate knowledge spillovers, which contribute to productivity improvements in nearby regions. For example, companies headquartered in central Tokyo may have expanded their operations to surrounding prefectures, such as Saitama, Chiba, and Kanagawa. These peripheral areas may have benefited from enhanced access to specialized labor and business networks. The sustained increase in Moran's I reflects a broader diffusion of agglomeration benefits.

Agglomeration shadows, where concentrated industrial development in core regions inhibits growth in neighboring areas (Fujita et al. 1999), may also be present. In the manufacturing sector, the decrease in Moran's I in the early 2000s may reflect these effects. As offshoring weakened the major manufacturing hubs, existing clusters may have become fragmented, and peripheral industrial bases may have deteriorated. For instance, strong industrial concentrations in regions, such as Aichi, Shizuoka, and Osaka, may have constrained the development of other areas through resource competition.

Among non-manufacturing areas, the rising dominance of the Tokyo metropolitan area may have placed pressure on regional economies. The agglomeration of finance, IT, and professional services in Tokyo may have undermined the competitiveness of similar sectors in provincial cities, thereby producing an agglomeration shadow. Consequently, productivity gaps between urban centers and peripheral areas may have widened, suppressing broader regional growth.

These observations reinforce the relevance of spatial modeling in explaining regional productivity dynamics. Section 4 applies spatial econometric models to quantify the influence of borrowed size effects and agglomeration shadows on RTFP.

3.2 Testing for region-specific effects

Accounting for unobserved region-specific heterogeneity is vital for spatial econometric analysis at the prefectural level. Time-invariant characteristics, such as geographic conditions, industrial composition, labor market structures, and local policy environments, may systematically influence RTFP. If such heterogeneity is ignored, the model estimates may suffer from omitted variable biases and inconsistencies.

To address this issue, this study evaluates whether fixed effects (FE) or random effects (RE) should be adopted in a panel data framework. Specifically, an F test for individual effects and the Hausman test for model consistency were conducted to guide the specification choice. These tests help determine whether unobserved regional differences should be treated as correlated with explanatory variables (FE) or as uncorrelated (RE), thereby improving estimation validity. The results are summarized in Table 3.

In the manufacturing and non-manufacturing sectors, F-statistics are highly significant, rejecting the null hypothesis of homogeneous regional effects. This confirms the presence of non-negligible cross-sectional heterogeneity and supports the need for region-specific effects.

The Hausman test results confirmed this conclusion. In both sectors, the test statistics are sufficiently large to reject the null hypothesis that the RE model provides consistent estimates. This outcome indicates that unobserved regional characteristics are likely to correlate with the explanatory variables, thereby validating the use of the FE model.

Overall, the test results strongly favor the fixed-effects specification. Therefore, Eqs. (4) and (5) are estimated under the fixed-effects framework in the spatial econometric analysis presented in Sect. 4.

Table 3 Model specification test results: F test and Hausman test

	Manufacturing sector	Non-manufacturing sector
F test	72.59 (0.00)	125.83 (0.00)
Hausman test	3812.47 (0.00)	102.33 (0.00)
Model selection results	FE	FE

Note: F test is the F-test statistic; Hausman test is the χ^2 -square value. Values in parentheses are p values. Model selection results represent a fixed-effects model (FE) or a random-effects model (RE)

Table 4 Estimation results of the SDM-FE model

	Manufacturing sector			Non-manufacturing sector		
Intercept	−0.122		(0.109)	−0.279	***	(0.050)
LQ	0.514	***	(0.026)	0.364	***	(0.042)
HC	0.712	***	(0.084)	0.053		(0.033)
TC	0.176	***	(0.019)	0.094	***	(0.008)
WLQ	−0.452	***	(0.028)	−0.368	***	(0.044)
WHC	−0.455	***	(0.105)	0.075	*	(0.042)
WTC	−0.158	***	(0.021)	−0.073	***	(0.008)
Rho	0.725	***	(0.021)	0.836	***	(0.014)
Lambda	N.A.			N.A.		
AIC	−1414.4			−2967.4		
LogL	716.2			1492.7		
LR test	623.5	***		1081.5	***	
LM test	4.871	**		16.210	***	
Wald test	1207.1	***		3534.4	***	

Note: standard errors are reported in parentheses

The LR test compares the SDM with a nested model, such as SAR or SEM

The LM test assesses remaining spatial error correlation

The Wald test assesses the joint significance of the model's coefficients

The spatial weight matrix used in the estimation is of the Queen type

All estimations were performed in R

AIC Akaike Information Criterion, LogL log-likelihood, N.A. not applicable (parameter not estimated in SDM model)

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4 Results

4.1 Estimation results

Tables 4 and 5 present the estimation results for the spatial econometric models. Table 4 presents the results from the SDM, and Table 5 displays the results from the SDM-SARAR.

First, the estimates from the SDM and SDM-SARAR indicate that many of the key explanatory variables are statistically significant at the 1% level and align with theoretical expectations. Specifically, LQ and TC exhibit consistently positive and significant effects in the manufacturing and non-manufacturing sectors. Conversely, in the non-manufacturing sector, the coefficients for HC and its spatial lag (WHC) are either statistically insignificant or exhibit weak effects, indicating limited contribution of human capital in clarifying productivity variation in this sector.

Subsequently, a comparison of the model fit using the Akaike Information Criterion (AIC) and log-likelihood values (LogL) revealed that the SDM-SARAR

Table 5 Estimation results of the SDM-SARAR-FE Model

	Manufacturing sector		Non-manufacturing sector	
Intercept	−1.288 ***	(0.435)	−0.180 ***	(0.041)
LQ	0.461 ***	(0.023)	0.374 ***	(0.042)
HC	0.752 ***	(0.075)	0.045	(0.033)
TC	0.158 ***	(0.017)	0.092 ***	(0.008)
WLQ	−0.144 ***	(0.043)	−0.376 ***	(0.043)
WHC	0.082	(0.153)	0.051	(0.040)
WTC	−0.038	(0.028)	−0.079 ***	(0.008)
Rho	−0.203 **	(0.076)	0.891 ***	(0.014)
Lambda	0.807 ***	(0.028)	−0.269 ***	(0.063)
AIC	−1417.9		−2980.9	
LogL	718.9		1500.4	
LR test	678.3 ***		1213.2 ***	

Note: standard errors are reported in parentheses

The LR test compared the SDM-SARAR model with the SDM to assess the need for spatial error dependence

The spatial weight matrix used in the estimation is of the Queen type

All estimations were performed in R

AIC Akaike Information Criterion, LogL log-likelihood

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

consistently outperformed the SDM, yielding lower AIC and higher LogL values. Furthermore, the likelihood ratio (LR) test, which evaluates the fit difference between the constrained (SDM) and unconstrained (SDM-SARAR) models, rejects the null hypothesis of equal model fit at the 1% significance level for both sectors. This indicates that the SDM-SARAR specification is statistically superior.

Residual diagnostics further supported this finding. The LM test results also detected significant residual spatial autocorrelation in the SDM for both sectors ($p < 0.05$, $p < 0.01$), supporting the appropriateness of including a spatially autocorrelated error term, as in the SDM-SARAR model. In addition, the Wald test confirmed the joint significance of all explanatory variables in the baseline SDM. Together, these results validate the SDM-SARAR model both theoretically and empirically.

Based on these findings, the estimated parameters of the SDM-SARAR were evaluated. The coefficient of LQ is positive and significant in both sectors, confirming that agglomeration contributes to productivity gains. In contrast, the spatial lag of agglomeration (WLQ) is negative and significant across both sectors, indicating a potential “agglomeration shadow” effect, whereby concentration of resources in neighboring regions suppresses local productivity.

In manufacturing, the coefficient for HC is positive and significant, but insignificant in non-manufacturing, implying that the role of human capital is limited in the

latter. The spatial lag of human capital (WHC) is insignificant in both sectors, indicating negligible potential spillovers of human capital from neighboring regions.

TC has a significant positive effect in both sectors. However, in the non-manufacturing sector, the spatial lag of transport capital (WTC) exerts a significantly negative effect, suggesting that spatial disparities in infrastructure investment may create interregional competition rather than shared benefits.

Finally, the spatial error component (λ) is positive and significant in manufacturing, but negative and significant in non-manufacturing. This asymmetry suggests complementary unobserved factors in the manufacturing industries and competitive interactions in the non-manufacturing industries.

Generally, the results suggest that while industrial agglomeration boosts productivity, excessive spatial concentration may induce negative externalities through the agglomeration shadow mechanism. The nature and strength of the borrowed size effects vary across sectors, with adverse spatial spillovers observed in the non-manufacturing sector. The negative spatial spillovers from transport capital highlight the need for careful regional coordination of infrastructure investment policies.

Although the SDM-SARAR model is theoretically flexible, it raises potential identification concerns. Particularly, when spatially lagged explanatory variables and spatial error terms are constructed using the same spatial weight matrix, it can be difficult to distinguish their impacts because of possible multicollinearity (Elhorst 2010).

To mitigate this risk, this study estimated the full SDM-SARAR specification using the `sacsarlm()` function in the `spatialreg` package in R by applying a maximum-likelihood estimation. Durbin terms were included for all explanatory variables, and the panel data structure ensured sufficient information richness across the spatial and temporal dimensions. The estimation was numerically stable for manufacturing and non-manufacturing models, and most of the key parameters were statistically significant.

These results suggest that appropriate model specifications, the use of rich spatial panel data, and the adoption of a Queen-type contiguity matrix contribute to stable estimation performance. Despite theoretical concerns regarding collinearity, empirical evidence supports the validity of the SDM-SARAR model in this context.

In conclusion, this specification successfully captures complex spatial interactions, including observable spillovers and unobserved spatial dependencies, without encountering convergence or instability. These findings underscore the practical utility of the model in analyzing regional productivity under spatial interdependence.

4.2 Robustness check

To assess the robustness of the spatial effects identified in the SDM-FE model, we re-estimated the model using an alternative spatial weight matrix based on the K-nearest neighbors (KNN) approach with $K=4$. While the original analysis employed a contiguity-based matrix defined by the Queen criterion, the KNN specification provides a complementary perspective by imposing a fixed number

Table 6 Robustness check of the SDM-FE Model

	Manufacturing sector		Non-manufacturing sector	
Intercept	−0.077	(0.115)	−0.319 ***	(0.055)
LQ	0.437 ***	(0.025)	0.477 ***	(0.042)
HC	0.626 ***	(0.076)	0.096 ***	(0.032)
TC	0.201 ***	(0.019)	0.087 ***	(0.008)
WLQ	−0.373 ***	(0.027)	−0.484 ***	(0.043)
WHC	−0.321 ***	(0.096)	0.048	(0.040)
WTC	−0.186 ***	(0.020)	−0.064 ***	(0.009)
Rho	0.716 ***	(0.021)	0.810 ***	(0.015)
Lambda	N.A.		N.A.	
AIC	−1347.9		−2840.9	
LogL	682.9		1429.5	
LR test	562.8 ***		933.8 ***	
LM test	0.914		0.834	
Wald test	1110.2 ***		2868.7 ***	

Note: standard errors are reported in parentheses

The LR test compares the SDM with a nested model, such as SAR or SEM

The LM test assesses remaining spatial error correlation

The Wald test assesses the joint significance of the model's coefficients

The spatial weight matrix used in the estimation was of the K-nearest neighbor type ($K=4$)

All estimations were performed in R

AIC Akaike Information Criterion, LogL log-likelihood, N.A. not applicable (parameter not estimated in SDM model)

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

of neighbors for each region, thereby mitigating the influence of spatial outliers and ensuring a balanced spatial representation across sparsely populated prefectures.

The estimation results under the KNN specification (Table 6) are broadly consistent with those obtained using the Queen-type matrix. The spatial lag coefficients for manufacturing and non-manufacturing sectors remain statistically significant, and the signs and magnitudes of the key explanatory variables—location quotient, human capital, and transport capital—are preserved across the model specifications, supporting the stability of the key relationships. This consistency suggests that the observed spatial structures are insensitive to the specifications of the spatial weight matrix.

Additionally, the LM test for residual spatial autocorrelation in the SDM-FE model under the KNN specification is not statistically significant for either sector. This result indicates that the SDM-FE model adequately captures the relevant spatial dependence, lending further support to the model's robustness.

Table 7 Robustness check of the SDM-SARAR-FE Model

	Manufacturing sector		Non-manufacturing sector	
Intercept	−0.517	(0.342)	−0.362 ***	(0.081)
LQ	0.423 ***	(0.023)	0.475 ***	(0.042)
HC	0.664 ***	(0.074)	0.098 ***	(0.032)
TC	0.188 ***	(0.018)	0.088 ***	(0.008)
WLQ	−0.211 ***	(0.049)	−0.483 ***	(0.043)
WHC	0.058	(0.168)	0.059	(0.044)
WTC	−0.126 ***	(0.028)	−0.061 ***	(0.009)
Rho	0.111	(0.129)	0.786 **	(0.033)
Lambda	0.676 ***	(0.065)	0.088	(0.096)
AIC	−1347.2		−2839.8	
LogL	683.6		1429.9	
LR test	607.6 ***		1072.2 ***	

Note: standard errors are reported in parentheses

The LR test compared the SDM-SARAR model with the SDM to assess the need for spatial error dependence

The spatial weight matrix used in the estimation was of the K-nearest neighbor type ($K=4$)

All estimations were performed in R

AIC Akaike Information Criterion, LogL log-likelihood

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We also estimated a more flexible spatial specification using the SDM-SARAR-FE model, which includes spatial lag and spatial error components (Table 7). The results suggest partial asymmetry: in manufacturing, the spatial error parameter (Lambda) is significant, while the spatial lag parameter (Rho) is not; in contrast, in non-manufacturing, Rho is significant, but Lambda is not. This divergence is not unexpected given the increased complexity and flexibility of the SDM-SARAR specification and may reflect sector-specific spatial spillover mechanisms, such as supply chain networks in manufacturing and service diffusion in non-manufacturing. Importantly, the main explanatory variables and spatial lag

Table 8 Impact effects from the SDM-SARAR Model

	Effect type	LQ	HC	TC
Manufacturing sector	Direct	0.455	0.748	0.157
	Indirect	−0.147	0.076	−0.035
	Total	0.308	0.824	0.122
Non-manufacturing sector	Direct	0.373	0.045	0.092
	Indirect	−0.079	0.049	−0.078
	Total	0.294	0.094	0.014

Note: this table reports the estimated impacts derived from the Spatial Durbin SARAR (SDM-SARAR) model

terms remain statistically significant and directionally stable, indicating that the core findings of this study are robust to alternative spatial specifications.

Overall, these robustness checks confirm the stability and reliability of the main results, reinforcing the empirical credibility and theoretical relevance of the identified borrowed size and agglomeration shadow effects.

4.3 Impact assessment

Based on the estimation results of the SDM-SARAR model (Table 5), the direct, indirect, and total effects of LQ, HC, and TC on RTFP were calculated using Eq. (6) (LeSage and Pace 2009). Table 8 presents the decomposed impact measures for the manufacturing and non-manufacturing sectors.

We first examined the manufacturing sector. Regarding direct effects, industrial agglomeration has a significantly positive impact, indicating that local clustering contributes to improvements in RTFP by facilitating technological spillovers and production efficiency. Human capital also exerts a positive and significant direct effect, with a magnitude exceeding that of agglomeration, highlighting the critical role of skilled labor in driving productivity. Transport capital similarly shows a significant direct effect, indicating that local infrastructure investment enhances production logistics and input accessibility.

Regarding indirect effects, industrial agglomeration exerts a significant negative influence, indicating the presence of an agglomeration shadow—a dynamic in which neighboring industrial hubs siphon away labor, capital, and technological inputs, thereby suppressing local productivity. In contrast, human capital demonstrates a significant positive spillover, implying the existence of interregional labor market integration and knowledge diffusion. The indirect effect of transport capital is negative and significant, implying that infrastructure in neighboring regions may intensify competition for mobile inputs or markets.

In terms of total effects, agglomeration and human capital contribute positively overall. However, the negative spatial spillover from industrial agglomeration is particularly pronounced, underscoring the importance of accounting for external pressure when formulating regional industrial strategies.

Next, we consider the non-manufacturing sector. Direct effects show that industrial agglomeration has a positive but weaker impact than in manufacturing, implying that while clustering supports productivity, it does not produce the same scale of technological spillovers. Human capital has an insignificant direct effect, suggesting that labor quality is less central in non-skill-intensive activities. In contrast, TC has a strong and direct effect, signifying that infrastructure development facilitates access to services and consumer markets in this sector.

As for the indirect effects, industrial agglomeration exhibits a negative effect, consistent with agglomeration shadows attributable to consumer and labor migration to urban centers. The indirect effect of human capital is positive but not statistically significant, whereas transport capital shows a significantly negative spillover, indicating that investment in core areas may draw economic activity away from the surrounding regions.

Regarding the total effects, industrial agglomeration and TC contribute positively to the non-manufacturing sector; however, the role of human capital remains marginal. The agglomeration shadow appears to be more binding in the non-manufacturing sector, where local compensating mechanisms, such as endogenous knowledge accumulation, are weaker and exposure to interregional competition is greater.

In summary, the decomposition of the spatial effects revealed sector-specific spatial dynamics. In manufacturing, endogenous growth drivers and positive spillovers from skilled labor strengthen resilience to spatial competition. In contrast, non-manufacturing sectors appear vulnerable to negative spillovers, as productivity depends on internal demand and physical infrastructure rather than on localized innovation or skill accumulation.

4.4 Policy implications

The findings suggest that effective spatial policies should be sector-specific and spatially adaptive. Given the observed asymmetries in spillover effects between the manufacturing and non-manufacturing sectors, a uniform approach is unlikely to achieve balanced regional development. Instead, different strategies tailored to the spatial dynamics of each sector are required.

The direct positive effects of agglomeration and human capital in the manufacturing sector imply that regional productivity can be enhanced by reinforcing local clustering and skill accumulation. This underscores the need for industrial policies that foster geographical collaboration and innovation. Simultaneously, negative spillovers from neighboring industrial hubs and evidence of agglomeration shadow effects highlight the risk of overconcentration. Therefore, policies should aim to promote regionally distributed cluster networks rather than singular industrial poles.

The agglomeration shadow effects identified in this study are likely to reflect specific structural mechanisms in Japan. For example, intensified interregional competition for skilled labor may drain talent from peripheral areas, whereas market siphoning effects concentrate demand in urban centers at the expense of nearby regions. Additionally, technological dependence on core regions may inhibit the development of independent innovation capabilities in less-developed prefectures. These mechanisms suggest that shadow effects are not merely statistical artifacts but manifestations of unequal regional interdependencies that merit targeted policy interventions.

To support this, regional growth strategies should include investments in mid-sized regional cities, which can function as secondary cores and help retain human capital. Educational and labor policies must also emphasize interregional mobility and integration, building broader functional labor markets that facilitate efficient talent matching across space.

Regarding transport infrastructure, the finding of negative spatial spillovers in manufacturing suggests that mere expansion may unintentionally weaken peripheral productivity. Thus, infrastructure planning should prioritize connectivity and complementarities over scale, with an emphasis on regional accessibility, multimodal integration, and the minimization of resource diversion.

In the non-manufacturing sector, policy challenges are acute. Spatial competition is pronounced, and the benefits of local agglomeration are weaker. The results highlight the need for policies that decentralize economic activity and strengthen place-based service economies such as tourism, culture, and local entrepreneurship.

To reduce dependence on metropolitan centers, peripheral regions should be supported in developing locally rooted economic identities bolstered by improvements in institutional capacity, quality of life, and public services. As human capital has a limited impact on this sector, higher returns can be achieved through strategic infrastructure investment, market expansion policies, and demand-side interventions, such as boosting local consumption through income stabilization.

Moreover, the potentially adverse impacts of metropolitan infrastructure on contiguous areas underscore the importance of evaluating the unintended spatial consequences of major public investments. Transport policies should be designed with a regionally differentiated lens to prevent the displacement of economic activity and strengthen economic resilience in non-core regions.

These implications reinforce the need for a spatially nuanced, sectorally targeted, and equity-conscious policy framework. Balancing growth with interregional fairness requires an integrated perspective that recognizes the opportunities and constraints created by spatial interdependence.

5 Conclusion

This study employed spatial econometric models to examine the influence of geographic and interregional factors on RTFP, focusing on the differences between the manufacturing and non-manufacturing sectors. Using panel data for 47 Japanese prefectures, the RTFP was calculated based on consistent measures of output and input and analyzed with respect to industrial specialization, human capital, and transport capital. By applying an SDM with spatial error components (SDM-SARAR), this study identified direct (local) and indirect (spillover) effects that are typically overlooked by conventional models.

A key insight from this analysis is that spatial interdependencies manifest differently across sectors and shape productivity outcomes through distinct spillover mechanisms. Industrial agglomeration exerts a positive effect on RTFP in both sectors, while negative spillovers from neighboring regions, interpreted as agglomeration shadow effects, are also statistically significant. Broadly, human capital contributes positively to productivity, particularly in the manufacturing sector. In contrast, the effect of transport capital is modest and occasionally negative, especially through spatial spillovers.

From theoretical and methodological perspectives, this study contributes to the literature by operationalizing the concepts of borrowed size and agglomeration shadow within a unified econometric framework. The SDM-SARAR model enables the simultaneous estimation of spatial spillovers and latent interregional dependencies, offering a comprehensive overview of how localized and external forces jointly shape regional productivity. This approach also advances empirical

spatial analysis by incorporating sectoral distinctions into the assessment of spatial externalities.

Several directions for future research must be explored. First, the mechanisms underlying spatial spillovers warrant further investigation. Although the SDM-SARAR model improves estimation by capturing spatial error dependence, future studies should consider identifying channels of interregional influence, namely, knowledge diffusion, labor mobility, and institutional interactions. Specifically, the conditions under which agglomeration shadows emerge, such as interregional competition for skilled labor or asymmetric innovation capacity, should be explored in detail.

Second, temporal dynamics should be incorporated. As the impacts of agglomeration, human capital, and infrastructure may evolve over time, dynamic spatial models are required to capture lagged adjustments and intertemporal spillover effects. Such extensions would offer a complete picture of regional development trajectories.

Third, a simulation-based policy analysis can enhance the practical relevance of spatial econometric findings. By modeling the potential effects of changes in industrial structure, educational attainment, or infrastructure investment, future studies could provide quantitative policy scenarios to inform spatial planning and regional development strategies.

While the present study provides robust empirical evidence of agglomeration shadow effects across Japanese prefectures, specific transmission mechanisms, such as labor competition, market centralization, and technological dependence, cannot be fully identified within the constraints of a macro-level spatial econometric framework. Addressing these channels requires complementary research designs, including case studies and qualitative institutional analyses, which fall outside the scope of this study. Nevertheless, the findings presented here lay an empirical foundation for future investigations by highlighting where and for which sectors shadow effects are most pronounced.

Despite the context-specific nature of the Japanese case, future studies should examine whether these findings can be generalized to other national or subnational settings, especially those with different institutional structures or spatial hierarchies, to assess the broader relevance of the borrowed size and agglomeration shadow frameworks proposed in this study.

Although the current analysis was limited to the pre-COVID-19 period due to data availability, the observed spatial patterns were consistent across multiple model specifications and weighting schemes. This stability suggests that the identified mechanisms, borrowed size and agglomeration shadow, are likely to hold beyond the study period.

Future studies should extend this framework to the post-2018 period, particularly to explore how regional productivity responded to the disruptions and recovery trajectories of the COVID-19 pandemic. Such extensions will provide deeper insights into the resilience and adaptability of Japan's regional economies under unprecedented conditions.

6 Data sources

The study data were accessed from publicly available sources (published in Japanese):

- Regional-Level Japan Industrial Productivity Database (R-JIP)
Research Institute of Economy, Trade, and Industry (RIETI)
<https://www.rieti.go.jp/jp/database/r-jip/>
- Employment Status Survey
Statistics Bureau of Japan
<https://www.stat.go.jp/data/shugyou/kako/index.html>
- Basic Survey on Wage Structure
Ministry of Health, Labour and Welfare
<https://www.mhlw.go.jp/toukei/list/chinginkouzou.html>
- Social Capital Stock Estimates
Cabinet Office of Japan
https://www5.cao.go.jp/keizai2/ioj/result/ioj_data.html

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Data availability Datasets used and/or analyzed in this study are available from the corresponding author upon request.

Declarations

Conflict of interest The author has no conflicts of interest to declare.

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